

Topic P2

Thermal physics

Concept guide · Worked examples · Cambridge-style practice · Fully explained answers

What this guide teaches

How the particle model explains solids, liquids and gases; gas pressure and volume; temperature scales and thermometers; thermal expansion; specific heat capacity; changes of state; evaporation; and conduction, convection and infrared radiation.

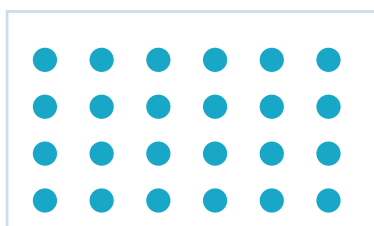
Syllabus map

P2.1.1	States of matter
P2.1.2	Particle model
P2.1.3	Gases and the absolute scale
P2.2.1	Thermal expansion
P2.2.2	Specific heat capacity
P2.2.3	Melting, boiling and evaporation
P2.3.1	Conduction
P2.3.2	Convection
P2.3.3	Radiation

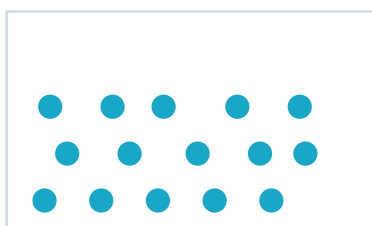
How to study: explain each observation using particles, cover the worked examples, and then complete the practice paper without looking at the solutions.

1. States of matter and the particle model

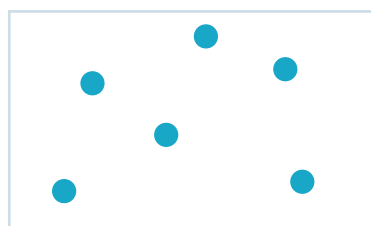
Matter is modelled as tiny particles in continuous random motion. The particles themselves do not expand when a substance is heated; their average kinetic energy and separation can change.



solid



liquid



gas

Solid: particles are closely packed in a regular arrangement and vibrate about fixed positions. Strong attractive forces give a fixed shape and volume. **Liquid:** particles remain close together but move past one another, so the liquid flows and has a fixed volume but no fixed shape. **Gas:** particles are widely separated, move rapidly and randomly, and fill their container.

Brownian motion

Irregular movement of visible smoke particles provides evidence for the random motion of much smaller, invisible air molecules. The smoke particles change direction because collisions from different sides are unequal at each instant.

Examiner insight

Avoid saying that particles in a substance become larger when heated. The model explains expansion by an increase in average particle separation.

2. Gas pressure and temperature

Gas pressure is caused by particles colliding with the walls of their container and changing momentum. More frequent collisions or larger momentum changes produce a greater force on the walls and therefore greater pressure.

Constant temperature: $pV = \text{constant}$ so $p_1V_1 = p_2V_2$

If volume decreases at constant temperature, particles travel a shorter distance between wall collisions. Collision frequency increases, so pressure increases. The relationship is inverse, not linear.

If temperature increases at constant volume, particles gain kinetic energy, move faster and collide more frequently and with greater momentum change. Pressure increases.

Worked example: compressed gas

A gas occupies 240 cm^3 at 100 kPa . It is compressed at constant temperature to 150 cm^3 . $p_2 = p_1V_1/V_2 = 100 \times 240/150 = 160 \text{ kPa}$.

Absolute temperature

The kelvin scale begins at absolute zero, the lowest possible temperature. For IGCSE calculations use:

$$T / \text{K} = \theta / ^\circ\text{C} + 273$$

Worked example: scales

$25\text{ }^{\circ}\text{C} = 25 + 273 = 298\text{ K}$. Conversely, $228\text{ K} = 228 - 273 = -45\text{ }^{\circ}\text{C}$. A temperature change of 1 K has the same size as a change of $1\text{ }^{\circ}\text{C}$.

3. Temperature, thermometers and expansion

Temperature indicates how hot an object is and is related to the average kinetic energy of its particles. Thermal energy flows spontaneously from higher temperature to lower temperature until thermal equilibrium is reached.

A thermometer uses a thermometric property that changes predictably with temperature, such as the volume of a liquid or the resistance of a thermistor. Two fixed points allow a scale to be calibrated.

Thermometer design

Change	Effect
narrower capillary	greater sensitivity
larger bulb	greater sensitivity but slower response
thinner bulb wall	faster response
longer scale / wider range	can measure a greater temperature interval

Thermal expansion occurs because heating increases particle vibration or motion and usually increases average separation. Most solids, liquids and gases expand when heated; gases generally expand most.

Applications and consequences

Expansion joints prevent bridges and railway tracks from buckling. A bimetallic strip bends because its two metals expand by different amounts. Liquid expansion is used in liquid-in-glass thermometers.

4. Specific heat capacity

Specific heat capacity c is the energy required to raise the temperature of 1 kg of a substance by 1 °C (or 1 K). A high value means more energy is needed for the same mass and temperature rise.

$$\Delta E = mc\Delta T$$

Use the temperature change ΔT , not the final temperature. The SI unit of c is J/(kg °C) or J/(kg K). Keep mass units consistent with the value of c .

Worked example: heating water

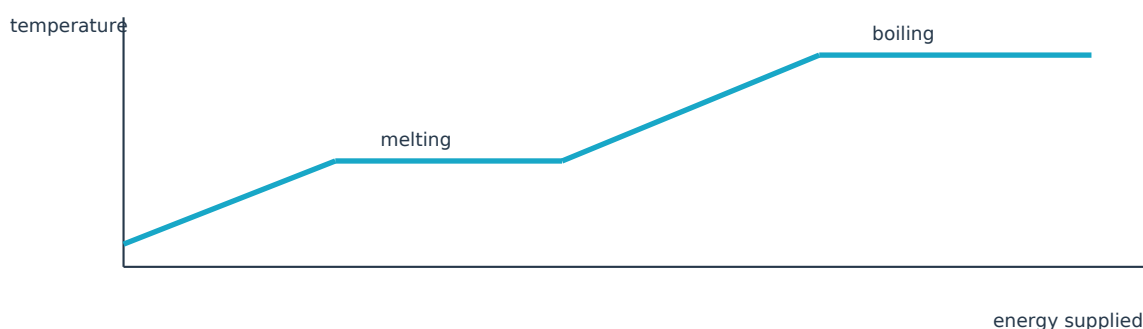
A 0.80 kg sample of water warms from 18 °C to 43 °C. For water, $c = 4200 \text{ J/(kg °C)}$. $\Delta T = 25 \text{ °C}$.
Energy transferred = $0.80 \times 4200 \times 25 = 84\,000 \text{ J}$.

Worked example: heater power

A 300 W heater transfers 36 000 J. From $P = E/t$, time = $36\,000/300 = 120 \text{ s}$. If energy is lost to the surroundings, the measured heating time is longer.

5. Changes of state and evaporation

During melting and boiling, energy supplied increases the separation and potential energy of particles rather than their average kinetic energy. The temperature remains constant until the change of state is complete.



Melting and freezing occur at the same temperature for a pure substance. Boiling occurs throughout a liquid at a fixed temperature when bubbles of vapour form. Evaporation occurs only at the surface and can happen below the boiling point.

Evaporation and cooling

The most energetic surface molecules are most likely to escape. The molecules left behind have a lower average kinetic energy, so the liquid cools. Evaporation is faster with higher temperature, larger surface area, greater air movement and lower humidity.

Common misconception

Boiling and evaporation are not interchangeable. Boiling occurs throughout the liquid at a fixed temperature; evaporation occurs at the surface over a range of temperatures.

6. Conduction

In solids, vibrating particles transfer energy to neighbours. Metals conduct especially well because mobile electrons move through the structure and transfer energy rapidly by collisions. Insulators have no mobile electrons and transfer energy more slowly.

Everyday application

A saucepan base is metal to conduct energy quickly to the food. Its handle is plastic or wood to reduce conduction to the hand. Trapped air in foam, wool and double glazing is a good insulator because air conducts poorly and cannot circulate easily.

7. Convection

Convection occurs in fluids. Heating causes a region of fluid to expand and become less dense. It rises, while cooler, denser fluid sinks to replace it. This circulation forms a convection current.

Examiner insight

Do not say that warm particles themselves “rise because heat rises”. State the linked sequence: expansion, lower density, upward movement, replacement by cooler denser fluid.

8. Infrared radiation

All objects emit and absorb infrared radiation. No medium is required, so radiation can transfer energy through a vacuum. Hotter objects emit more radiation per second.

Dull black surfaces are good absorbers and good emitters. Shiny light surfaces are poor absorbers and poor emitters but good reflectors of infrared radiation.

Worked reasoning

A dull black can cools faster than a shiny can at the same temperature because it is a better emitter. Under a radiant heater, the black surface warms faster because it is also a better absorber.

Reducing thermal transfer

- Vacuum insulation prevents conduction and convection.
- Foam or trapped air reduces conduction and prevents large convection currents.
- Shiny surfaces reflect infrared and are poor emitters.
- A lid reduces convection and evaporation from the surface.

Exam technique from the 2026 evidence

The supplied June 2026 papers assess thermal physics through calculations, particle explanations, applications and practical comparisons. Precision of language matters.

High-value habits

1. Refer to particles, collisions and momentum change when explaining gas pressure. 2. Use ΔT in specific heat capacity calculations. 3. Distinguish absorption from emission. 4. Explain convection using density differences. 5. Convert $^{\circ}\text{C}$ and K using 273 in the correct direction.

Frequent errors

- Treating pressure and volume as directly or linearly related.
- Saying that particles expand when matter is heated.
- Calling a shiny surface a good emitter.
- Using final temperature instead of temperature change.
- Writing 273°C instead of -273°C for absolute zero.

Five-minute retrieval

- Sketch particle arrangements for a solid, liquid and gas.
- Explain gas pressure in one sentence using collisions.
- Write the three equations in this topic from memory.
- Compare boiling and evaporation.
- Explain one method of reducing each transfer process.

Practice Paper

Topic P2

Thermal physics**Instructions**

Answer all questions. For multiple-choice questions, select one option. Show all working for calculations. Write explanations on the lines provided. Use $T/K = \theta/^{\circ}\text{C} + 273$.

Candidate name:

Class / date:

Total marks: 50

Suggested time: 60 minutes. A separate worked solution section follows the answer booklet.

Section A · Multiple choice

Choose the best answer for each question.

1. Which statement describes particles in a liquid? [1]
- A They vibrate about fixed positions only.
 - B They are far apart and move independently.
 - C They are close together and move past one another.
 - D They have no attractive forces between them.
2. A gas is compressed at constant temperature. Why does its pressure increase? [1]
- A Particles become larger.
 - B Particles collide with the walls more frequently.
 - C Particles gain gravitational potential energy.
 - D The number of particles increases.
3. What is 35 °C on the kelvin scale? [1]
- A 238 K
 - B 273 K
 - C 308 K
 - D 335 K
4. Which change makes a liquid-in-glass thermometer more sensitive? [1]
- A a wider capillary
 - B a narrower capillary
 - C a thicker bulb wall
 - D a smaller scale
5. Which equation is used to calculate energy transferred when temperature changes? [1]
- A $E = mc\Delta T$
 - B $E = m/c\Delta T$
 - C $E = mc/T$
 - D $E = c\Delta T/m$
6. Why does temperature remain constant while a pure solid melts? [1]
- A No energy enters the solid.
 - B Energy increases particle separation rather than average kinetic energy.
 - C Particles stop moving.
 - D The solid loses internal energy.

7. Which change increases the rate of evaporation? [1]
- A decreasing surface area
 - B decreasing air movement
 - C increasing humidity
 - D increasing temperature
8. Why are metals good thermal conductors? [1]
- A They contain mobile electrons.
 - B Their particles do not vibrate.
 - C They contain convection currents.
 - D They absorb all infrared radiation.
9. What causes warm water to rise in a convection current? [1]
- A It becomes more dense.
 - B It becomes less dense.
 - C Its particles become larger.
 - D Gravity no longer acts on it.
10. Which surface is the best emitter of infrared radiation? [1]
- A dull black
 - B shiny black
 - C dull white
 - D shiny silver

Section B · Structured questions

11 A gas occupies 300 cm^3 at a pressure of 120 kPa . Its temperature remains constant.

- (a) Explain, in terms of particles, how the gas produces pressure on the walls of its container. [3]

- (b) The gas is compressed to 180 cm^3 . Calculate its new pressure. [3]

- (c) The gas is then heated at constant volume. State and explain what happens to its pressure. [2]

[Total: 8]

12 A 0.60 kg block has a specific heat capacity of $900 \text{ J/(kg } ^\circ\text{C)}$. Its temperature increases from $20 ^\circ\text{C}$ to $70 ^\circ\text{C}$.

- (a) Calculate the temperature change of the block. [1]

- (b) Calculate the energy transferred to the block. [3]

(c) A 250 W heater provides this energy. Calculate the minimum heating time. [2]

(d) State two reasons why the actual heating time is greater. [2]

[Total: 8]

13 A student heats a pure solid until it melts and then boils.

(a) State what happens to the temperature while the solid is melting. [1]

(b) Explain, using particles, why energy is still required during melting. [2]

(c) Explain why evaporation causes a liquid to cool. [3]

(d) State two differences between boiling and evaporation. [2]

[Total: 8]

14 A metal spoon and a wooden spoon are placed in hot soup.

(a) Explain why the metal spoon becomes hot more quickly than the wooden spoon. [3]

(b) Describe how a convection current forms in the soup. [3]

(c) Explain why convection cannot occur in the solid spoons. [2]

[Total: 8]

15 A student compares the absorption of infrared radiation by a dull black surface and a shiny silver surface.

(a) Predict which surface shows the greater temperature increase. [1]

(b) Explain your prediction. [2]

(c) Describe an experiment to compare the two surfaces. Include the apparatus, measurements and variables that must be controlled. [3]

(d) State two ways to improve the reliability of the result.

[2]

[Total: 8]

Worked solutions

Remove or fold back this section while completing the practice paper. Mark only after finishing the full attempt.

Marking routine

Award marks for correct physics and method, not only the final number. Identify the first error as knowledge, equation choice, rearrangement, unit conversion, particle explanation or communication.

Solutions · Multiple choice

1. C - liquid particles remain close but can change neighbours.
2. B - compression reduces the distance between wall collisions.
3. C - $35 + 273 = 308 \text{ K}$.
4. B - a narrower capillary gives a larger length change for the same volume expansion.
5. A - $E = mc\Delta T$.
6. B - energy separates particles and increases potential energy.
7. D - higher temperature gives more molecules enough energy to escape.
8. A - mobile electrons transfer energy rapidly through a metal.
9. B - expansion makes the warm water less dense.
10. A - a dull black surface is the best emitter.

Solutions · Structured questions

11(a) [3] Particles move randomly [1], collide with the walls [1], and change momentum / exert a force on the walls [1].

11(b) [3] $p_1V_1 = p_2V_2$ [1]. $p_2 = 120 \times 300/180$ [1] = 200 kPa [1].

11(c) [2] Pressure increases [1] because particles move faster and collide more frequently / with greater momentum change [1].

12(a) [1] $\Delta T = 70 - 20 = 50$ °C.

12(b) [3] $\Delta E = mc\Delta T$ [1] = $0.60 \times 900 \times 50$ [1] = 27 000 J [1].

12(c) [2] $t = E/P$ [1] = $27\,000/250 = 108$ s [1].

12(d) [2] Any two: energy heats the container; energy is transferred to the air; heater power is not fully transferred to the block [1 each].

13(a) [1] The temperature remains constant.

13(b) [2] Energy is used to overcome attractive forces / increase particle separation [1], increasing particle potential energy rather than average kinetic energy [1].

13(c) [3] More energetic surface molecules escape [1]; the remaining molecules have lower average kinetic energy [1]; therefore the liquid temperature decreases [1].

13(d) [2] Boiling occurs throughout the liquid at a fixed temperature [1]; evaporation occurs at the surface and can occur below the boiling point [1].

14(a) [3] Metal contains mobile electrons [1] that gain energy near the hot end [1] and transfer it rapidly by collisions through the metal [1].

14(b) [3] Warm soup expands and becomes less dense [1], rises [1], and cooler denser soup sinks to replace it, forming a circulation [1].

14(c) [2] Particles in a solid cannot move from place to place [1], so bulk movement / a convection current cannot form [1].

15(a) [1] The dull black surface.

15(b) [2] Dull black is a better absorber of infrared radiation [1], so it receives more energy per second and warms faster [1].

15(c) [3] Place identical black and silver surfaces the same distance from an infrared lamp [1]. Attach identical thermometers and record temperature change over the same time [1]. Control starting temperature, distance, exposed area and lamp power [1].

15(d) [2] Repeat and calculate a mean [1]; collect readings over a longer interval / use digital probes or data logging [1].

Self-assessment

Mark each answer, circle the first incorrect step, and label the cause: knowledge, equation choice, rearrangement, unit conversion, particle explanation or communication. Revise the label that occurs most often.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.